

LOOKING AT CLADOGRAMS

To do a cladistic analysis, the different groups of organisms are scored on a set of characters that are either primitive or derived. Their distribution is not always as straightforward as is shown in the diagrams below. Often the resulting cladogram can be constructed in a number of different ways and taxonomists have to choose between them. To do this they adopt the principle of parsimony – they choose the cladogram involving the least number of steps or character transformations to explain the observed relationships between the groups.



< MILKY DIET

All mammals possess mammary glands. Unique to the class Mammalia, this feature is synapomorphic at this taxonomic level. To find further familial relationships within the class, characters that are synapomorphic at familial level are used.

CHARACTER	CANID	BEAR	SEAL	FUR SEAL & SEA LION	WALRUS
Feeding young on milk	1	1	1	1	1
Short tail	0	1	1	1	1
Forelimbs modified into flippers	0	0	1	1	1
Very flexible spine	0	0	1	1	1
Hind limbs turn forwards under body	0	0	0	1	1
Presence of tusks	0	0	0	0	1

< CHARACTER SET

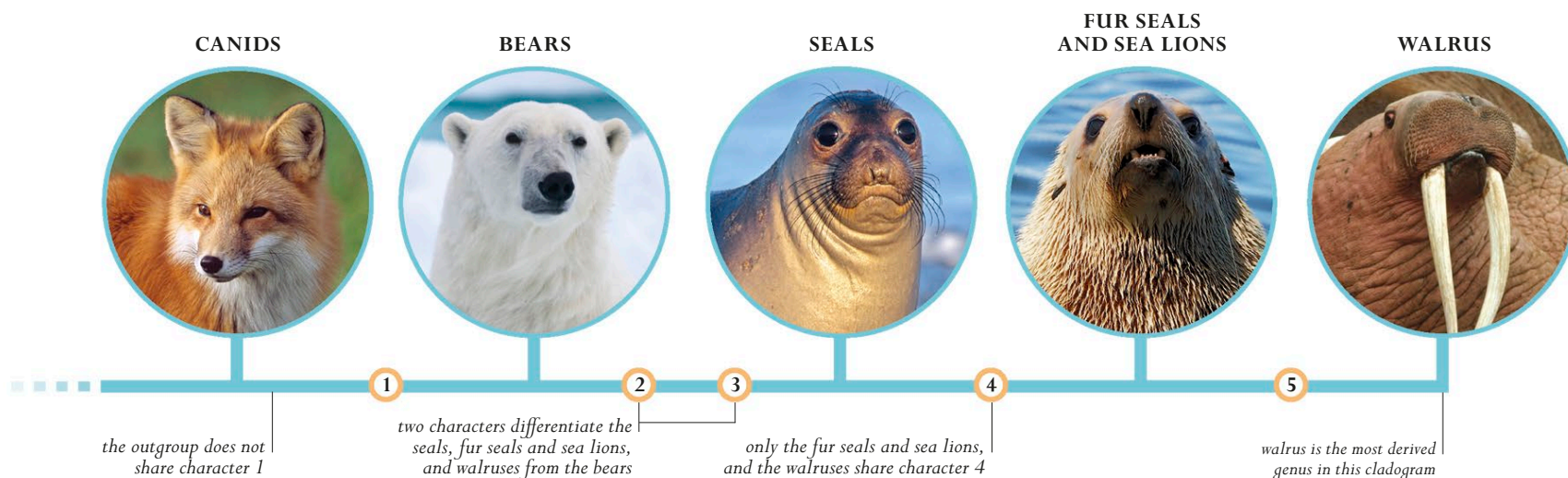
Most modern cladograms are based on genetics using DNA codes. The codes that were used to generate the cladogram below have been replaced with more familiar morphological descriptions in this character set. One character, feeding young on milk, is shared by all the groups shown; some characters are found in only some of the groups; and one character, the presence of tusks, is unique to the walrus.

✓ CLADOGRAM

In this cladogram the canids are considered the most primitive group, known as the outgroup, and the walrus is the most derived. All characters on the cladogram are shared by the groups to the right of each number – for example, a short tail is shared by the bears, seals, fur seals and sea lions, and the walrus.

KEY

- | | |
|---|---------------------|
| 0 | primitive character |
| 1 | derived character |



SHORT TAIL

Bears, seals, fur seals and sea lions, and the walrus all have short tails (character 1). Canids, however, display the primitive condition and have long, bushy tails. Character 1 is therefore a synapomorphic character shared by all the carnivore families shown except the canids.



FLIPPERS

Among carnivores, the modified limbs of seals, fur seals and sea lions, and the walrus are unique. Character 2 (forelimbs modified into flippers) is therefore synapomorphic at this level, suggesting these three groups are more closely related to one another than to the bears.



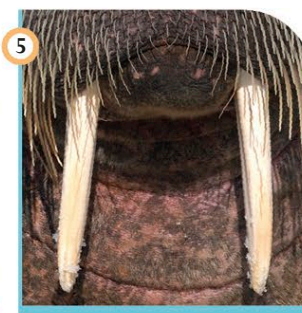
FLEXIBLE SPINE

Character 3 (flexible spine) operates at the same level as character 2, further supporting the relationship indicated by possession of flippers. The more synapomorphic characters that appear at a particular level, the more convincing the suggested relationship becomes.



SUPPORTING LIMBS

In both the fur seals and sea lions and the walrus, the pelvic girdle can be rotated to allow the hind limbs to aid locomotion on land. This suggests they share a more recent common ancestor with each other than they do with the seals, which are far less mobile when out of water.



TUSKS

This is a unique character (autapomorphy) of the walrus that reveals nothing about its relationship to other mammal families. The same is true of the character "feed young on milk" – present in all the groups, it provides no clues as to how they interrelate, so it is not plotted on the cladogram.